

TP4056

TP4056 — 1 A Li-Ion Battery Charger

Manufacturer: Top Power Package: SOP-8

General Description

The TP4056 is a complete constant-current / constant-voltage linear charger for single-cell lithium-ion batteries, with thermal regulation and a programmable charge current up to 1 A.

Features

- Programmable charge current up to 1 A
- Input voltage range 4.5 V to 8 V
- Constant-current / constant-voltage operation
- Thermal regulation, automatic recharge
- Charge status outputs CHRG and STDBY

Pin Configuration

Pin assignment for the SOP-8 package (top view).

Pin	Name	Function
1	TEMP	Battery temperature sense input
2	PROG	Charge current program resistor
3	GND	Ground
4	VCC	Supply input 4.5 to 8 V
5	BAT	Battery connection
6	STDBY	Charge complete indicator, open drain
7	CHRG	Charging indicator, open drain
8	CE	Chip enable, active high

Absolute Maximum Ratings

Stresses beyond these values may cause permanent damage to the device.

Symbol	Parameter	Value	Unit
VCC	Input supply voltage	8.0	V
VBAT	Battery pin voltage	7.0	V
IBAT	Charge current	1.0	A
Tj	Operating junction temperature	125	degC

Electrical Characteristics

Typical values at TA = 25 degC unless otherwise noted.

Symbol	Parameter	Test condition	Min	Typ	Max	Unit
VCC	Input supply voltage	-	4.5	5.0	8.0	V
VFLOAT	Regulated output float voltage	-	4.158	4.200	4.242	V
IBAT	Charge current	RPROG = 1.2 kohm	900	1000	1100	mA
VTRIM	Trickle charge threshold	-	2.8	2.9	3.0	V

Thermal Characteristics

Symbol	Parameter	Value	Unit
RthJA	Thermal resistance junction-ambient	120	K/W

Package Dimensions

Outline drawing for the SOP-8 package. All dimensions in millimetres.

Dimension	Min	Nom	Max	Unit
A (body length)	4.80	4.90	5.00	mm
B (body width)	3.80	3.90	4.00	mm
e (lead pitch)	-	1.27	-	mm

Ordering Information

Order code	Package	Marking	Packing
TP4056	SOP-8	TP4056	Reel, 2500 pcs

Document revision 1.0. Values are typical at TA = 25 degC. Verify against the latest datasheet from Top Power before design release.